Function and Array Assessment

A medium-difficulty assessment testing concepts of array manipulation, function parameters, and return values.

SECTION B: Short Answer Questions

Q1. Write a function named 'findAverage' that takes an array of integers as a parameter and returns the average of the elements as a float. (5 Marks)

Q2. Explain the difference between passing an array to a function by value versus passing it by reference, and state which is the default in C++. (5 Marks)

SECTION C: Long Descriptive Questions

Q1. Design a function 'removeDuplicates' that accepts an array of integers, removes any duplicate elements in-place, and returns the new size of the array. Write the pseudocode and explain the time complexity of your approach. (10 Marks)

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

Short Questions - Grading Guide

Q1. Ideal Answer:
The function should initialize a sum variable, loop through the array to accumulate the total, and return the sum divided by the array length.
Rubric:
Award 3 marks for correct loop accumulation and 2 marks for correct division and return type.

Q2. Ideal Answer:
Passing by value copies the entire array, whereas passing by reference passes the memory address of the original array. In C++, arrays are passed by reference by default.
Rubric:
Award 3 marks for explaining the difference in memory usage and 2 marks for identifying the default C++ behavior.

Long Questions - Grading Guide

Q1. Ideal Answer:
The function uses a write-pointer to overwrite duplicates as it iterates through the sorted array, returning the write-pointer's final index. This in-place operation achieves O(N log N) time complexity if sorting is required, or O(N) if already sorted.
Rubric:
Award 5 marks for correct in-place shifting logic, 3 marks for accurate pseudocode, and 2 marks for the complexity analysis.